translationBridge

Service Description

Service ID: *"translationBridge"*

Abstract
This document describes the `translationBridge` service, version 1.0.0, produced by the TranslationManager support system of Arrowhead (documentation edition 5.3.0).

Contents

1 Documentation edition and implementation scope **3**

2 Overview **3**
2.1 Significant Prior Art . . . 3
2.2 How This Service Is Meant to Be Used . . . 3
2.3 Important Delimitations and Dependencies . . . 3

3 Service Interface **4**
3.1 operation discovery . . . 4
3.2 operation negotiation . . . 4
3.3 operation abort . . . 4

4 Information Model **5**
4.1 struct Address . . . 5
4.2 struct DeviceResponse . . . 5
4.3 struct ServiceDefinitionResponse . . . 5
4.4 struct ServiceInstanceInterfaceResponse . . . 5
4.5 struct ServiceInstanceResponse . . . 5
4.6 struct SystemResponse . . . 6
4.7 struct TranslationBridgeCandidate . . . 6
4.8 struct TranslationDiscoveryRequest . . . 6
4.9 struct TranslationDiscoveryResponse . . . 7
4.10 struct TranslationNegotiationRequest . . . 7
4.11 struct TranslationNegotiationResponse . . . 7
4.12 Primitives . . . 7

5 References **9**

6 Revision History **10**
6.1 Amendments . . . 10
6.2 Quality Assurance . . . 10

1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

2 Overview

This document describes the `translationBridge` service, which provides interface and data-model translation for a consumer: finding providers that become usable through a translation bridge, building the bridge and aborting it. It is produced by the TranslationManager system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

2.1 Significant Prior Art

Successor of the Arrowhead 4.x Translator and Gateway concepts, redesigned for Arrowhead 5 interface templates and data-model identifiers.

2.2 How This Service Is Meant to Be Used

DynamicServiceOrchestration normally drives translation (orchestration flag `ALLOW_TRANSLATION`, `enable.translation=true`): `discovery` with the candidates and the consumer's interface and data models, then `negotiation` for the chosen candidate. The consumer calls the returned bridge interface instead of the provider. A consumer may also use the service directly.

2.3 Important Delimitations and Dependencies

Translation needs interface and data-model translator systems registered in the cloud; the TranslationManager orchestrates them but does not translate itself.

3 Service Interface

This section describes the interfaces to the `translationBridge` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.



Figure 1: SysML-style block of the `translationBridge` service and its operations.

The following interface operations are available.

3.1 operation discovery (TranslationDiscoveryRequest) : TranslationDiscoveryResponse

Returns the candidates usable by the requester through translation, with a bridge identifier.

3.2 operation negotiation (TranslationNegotiationRequest) : TranslationNegotiationResponse

Builds the bridge to the chosen candidate and returns the bridge interface and token limits.

3.3 operation abort (Identifier)

Aborts a bridge of the requester. Returns 200 if aborted, 204 if there was nothing to abort. The bridge identifier fills the `{bridgeId}` placeholder of the registered path.

4 Information Model

Here, all data objects that can be part of the `translationBridge` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.12, which are used to represent things like hashes and identifiers.

4.1 struct Address

A network address with its type.

Field	Type	Description
type	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
address	String	One network address (host name, IPv4, IPv6 or MAC address).

4.2 struct DeviceResponse

A registered device.

Field	Type	Description
name	Name	Device name.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<Address>	Network addresses with their types.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.3 struct ServiceDefinitionResponse

A service definition.

Field	Type	Description
name	Name	Service definition name.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.4 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

Field	Type	Description
templateName	Name	Name of the interface template, snake_case (e.g. `generic_https`).
protocol	String	Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`.
policy	String	Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies.
properties	Map<String, Object>	Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`.

4.5 struct ServiceInstanceResponse

A registered service instance.

Field	Type	Description
instanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`, or policy instance identifier.
provider	SystemResponse	The provider system.
serviceDefinition	ServiceDefinitionResponse	The service definition.
version	Version	Version of the service instance.
expiresAt	DateTime	UTC expiry time; no value means no expiry.
metadata	Metadata	Free-form metadata (JSON object).
interfaces	List<ServiceInstanceInterface Response>	Service interfaces offered by the instance.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.6 struct SystemResponse

A registered system.

Field	Type	Description
name	Name	System name.
metadata	Metadata	Free-form metadata (JSON object).
version	Version	Version of the system.
addresses	List<Address>	Network addresses with their types.
device	DeviceResponse	The device hosting the system, if registered.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

4.7 struct TranslationBridgeCandidate

A candidate reachable through translation.

Field	Type	Description
serviceInstanceId	Identifier	Service instance identifier `<System>|<serviceDefinition>|<version>`.
interfaceTemplateName	String	Interface template of the candidate the bridge would translate to.

4.8 struct TranslationDiscoveryRequest

Discovery of translation bridges for the requester.

Field	Type	Description
candidates	List<ServiceInstanceResponse>	Candidate service instances (from service discovery).
operation	String	Operation the consumer wants to call.
interfaceTemplateNames	List<String>	Interface templates the consumer can use.
inputDataModelId	Identifier	Data model of the operation input expected by the consumer.
outputDataModelId	Identifier	Data model of the operation output expected by the consumer.

4.9 struct TranslationDiscoveryResponse

Candidates reachable through a translation bridge.

Field	Type	Description
bridgeId	Identifier	Identifier (UUID) of the translation bridge.
candidates	List<TranslationBridgeCandidate>	Store entries to create.

4.10 struct TranslationNegotiationRequest

Builds the bridge to the chosen candidate.

Field	Type	Description
bridgeId	Identifier	Identifier (UUID) of the translation bridge.
target	ServiceInstanceResponse	The chosen candidate service instance.
operation	String	QoS operation: `FILTER` or `SORT`.
interfaceTemplateName	String	Interface template the consumer will use.
inputDataModelId	Identifier	Data model of the operation input expected by the consumer.
outputDataModelId	Identifier	Data model of the operation output expected by the consumer.

4.11 struct TranslationNegotiationResponse

The established bridge.

Field	Type	Description
bridgeId	Identifier	Identifier (UUID) of the translation bridge.
bridgeInterface	ServiceInstanceInterfaceResponse	Interface of the bridge the consumer calls instead of the provider.
tokenExpiresAt	DateTime	Expiry of the bridge token.
tokenUsageLimit	Number	Number of uses allowed with the bridge token.

4.12 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
List	An *array* of a known number of items, each having type A.
Map	A set of key-value pairs.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
Object	Any structured value.
String	A chain of characters.
Version	Specifies a version, semantic versioning `major.minor.patch`.

5 References

[1] J. Delsing, “TranslationManager – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-TranslationManager-java-spring/documentation/v5.3.0/TranslationManager-v5.3.0-SysD.pdf.

[2] ——, “translationBridge HTTP/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, translationManager-v5.3.0-translationBridge-HTTP-IDD.pdf.

[3] ——, “translationBridge MQTT/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, translationManager-v5.3.0-translationBridge-MQTT-IDD.pdf.

6 Revision History

6.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)
3	2026-11-09	5.3.0	New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained.	Documentation maintenance

6.2 Quality Assurance

No.	Date	Version	Approved by
1		5.3.0	